

Assignment 4

1. The two numbers represented in signed 2's complement form are $P = 11101101$ and $Q = 11100110$. If Q is subtracted from P , then what will be the value obtained in signed 2's complement??
2. 11001, 1001, 111001 correspond to the 2's complement representation. Find out their equivalent decimal number??
3. In the circuit shown, diodes D_1 , D_2 and D_3 are ideal, and the inputs E_1 , E_2 and E_3 are "0 V" for logic '0' and "10 V" for logic '1'. What logic gate does the circuit represent? (Refer Fig. 1)

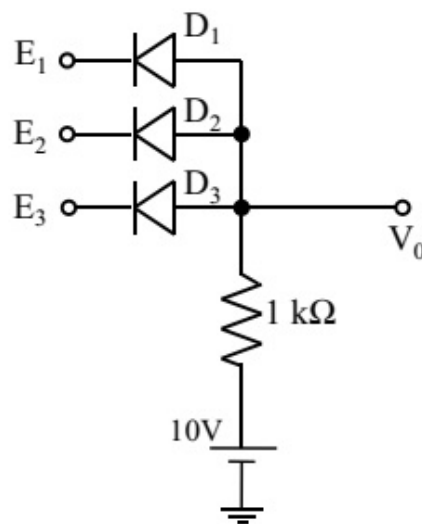


Fig. 1

4. A function $F(A,B,C)$ defined by three Boolean variables A , B and C when expressed as sum of products is given by $F = A'B'C' + A'BC' + AB'C'$ where A' , B' and C' are the complements of the respective variables. Find out product of sum for F ??
5. Find out the output of the combinational circuit given (Fig. 2)

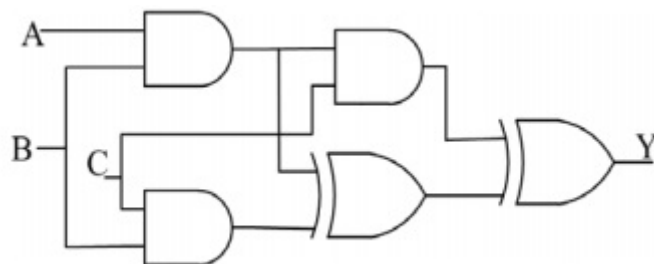


Fig. 2

6. For the output F to be 1 in the logic circuit shown, the input combination should be (Fig. 3)

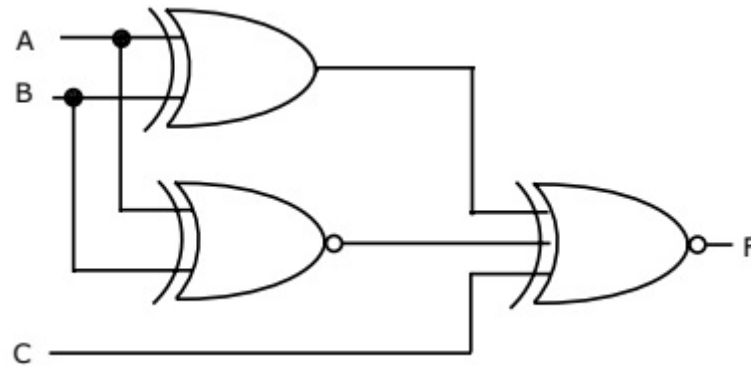


Fig. 3

7. A coil of 200 turns is wound uniformly over a wooden ring having a mean circumference of 600 mm and a uniform cross-sectional area of 500 mm^2 . If the current through the coil is 4.0 A, calculate (a) the magnetic field strength; (b) the flux density; (c) the total flux.
8. Calculate the magnetomotive force required to produce a flux of 0.015 Wb across an air-gap 2.5 mm long, having an effective area of 200 cm^2 .
9. The simple magnetic circuit shown has a cross-sectional area of 50 cm^2 and mean length of 2 m. The relative permeability of the core is 100. The coil has 250 turns and the flux produced is $100 \mu\text{Wb}$. Find reluctance of magnetic circuit and current flowing through coil. (Refer Fig. 4)

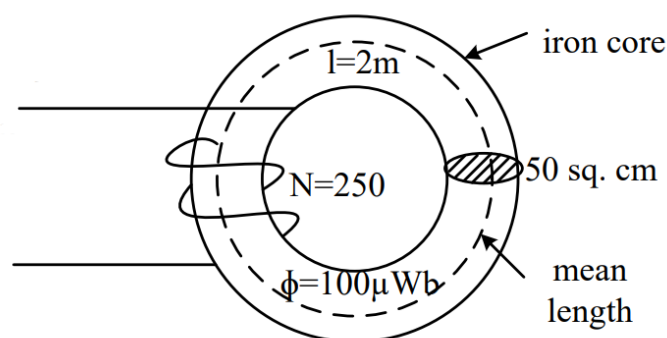


Fig. 4

10. An iron ring of mean circumference 50-cm has an air gap of 0.1-cm length and a winding of 300 turns. If the relative permeability of iron is 400 when a current of 1 ampere flows through the coil, find the flux density in the air gap. (Refer Fig. 5)

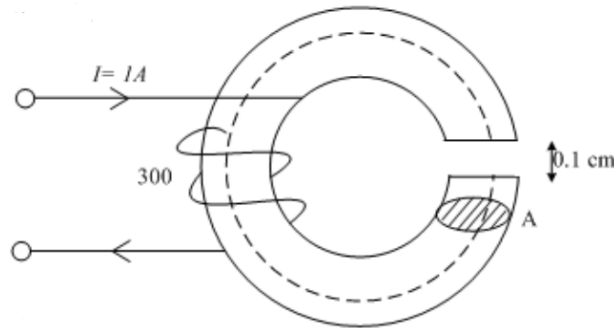


Fig. 5

11. Two identical coils A and B of 1000 turns each lie in parallel plane such that 80% of the flux produced by one coil links with other. If a current of 5 A flowing in coil A produces flux of 0.05 mWb, then what is the flux linking with coil B??
12. The mean length of a magnetic circuit is 40 cm with cross-sectional area 8 cm² and relative permeability of the core material being 200. When a magnetizing coil carries a current of 6 A, the magnetic flux produced in the core is 3 mWb. Find: (i) reluctance of the magnetic circuit (ii) flux density in the core (iii) magnetic field intensity (iv) number of turns of the coil.
13. A linear magnetic circuit has a flux linkage of 2 Wb-turn when a current of 20 A flows through its coil. What is the energy stored in the magnetic field of the coil?
14. A mild steel (relative permeability of 400) ring having a cross-sectional area of 500 mm² and a mean circumference of 400 mm has a coil of 200 turns wound uniformly around it. Calculate (a) The reluctance of the ring (b) The current required to produce a flux of 800 μ Wb in the ring.
15. A negative feedback system as shown in Fig. 6 has an amplifier of gain 100 with $\pm 10\%$ tolerance in the forward path, and an attenuator of value $9/100$ in the feedback path. The overall system gain is approximately:

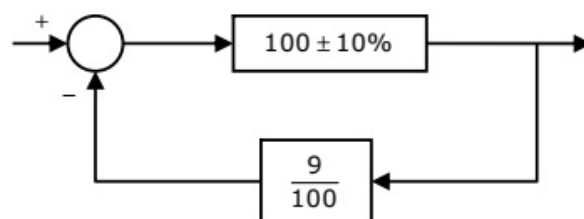


Fig. 6

16. For the block diagram shown in the Fig. 7, the transfer function $\frac{Y(s)}{R(s)}$ is

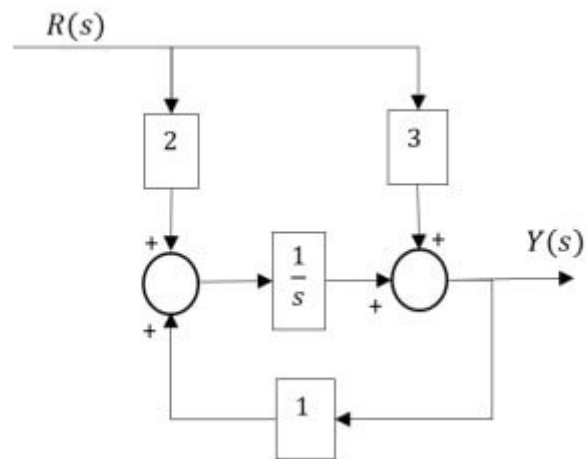


Fig. 7

17. The transfer function $\frac{V_2(s)}{V_1(s)}$ of the circuit shown below is (Refer Fig. 8)

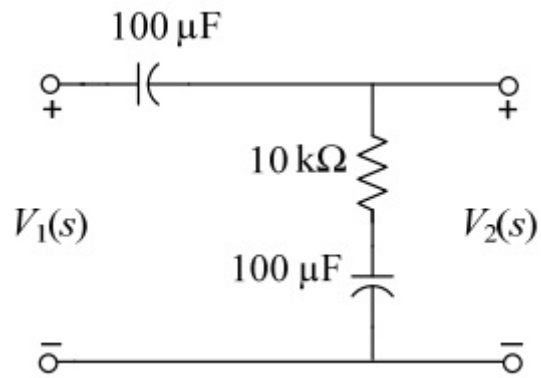


Fig. 8